35

Emerging Markets*

The world's emerging economies, home of 86 percent of the population, accounted for about 59 percent of global GDP in 2017 and are growing faster than the developed economies.[1] As emerging markets become more important to the global economy and to investors, sound methods are needed for analyzing and valuing companies and business units in these markets.

Chapters 26 and 27 discussed general issues related to forecasting cash flows, estimating the cost of capital in a foreign currency, and incorporating high inflation rates into cash flow projections. This chapter focuses on additional issues that arise in emerging markets, such as the potential for extreme economic contractions or unexpected government actions like asset appropriation. It is impossible to generalize about these risks, as they differ by country and may affect businesses in different ways. Academics, investment bankers, and industry practitioners subscribe to different methods and often make arbitrary adjustments based on intuition and limited empirical evidence.

For accurate valuation of companies in emerging markets, we recommend using a scenario discounted-cash-flow (DCF) approach as described in Chapter 16 to prepare multiple cash flow scenarios reflecting the outcomes of different risks that a company could face. These scenarios are each discounted and then weighted by probabilities assigned to each. You can supplement this method by comparing the results with two secondary approaches: a DCF valuation with a country risk premium built into the cost of capital and a valuation based on the multiples of comparable companies.

* The authors would like thank Andre Gaeta, Daniel Guzman, Paulo Guimaraes, Joao Lopes Sousa, and Barbara Castro for their contributions to this chapter.

[1] China's and India's shares of global GDP, at purchasing parity prices (PPP), were 19 and 8 percent, respectively, and 19 and 18 percent of population, respectively. International Monetary Fund, "GDP Based on PPP, Share of World," IMF DataMapper, imf.org.

WHY SCENARIO DCF IS MORE ACCURATE THAN RISK PREMIUMS

The most vigorously debated issue about valuing companies in emerging markets is whether to incorporate a country risk premium in the cost of capital. A common practice has been to add a country risk premium to the discount rate to account for the higher risks of operating in emerging markets.[2] Often, the premium is based on the government's borrowing rate relative to a benchmark, such as the borrowing rates for the U.S. government.

A major problem with this approach is that the riskiness of lending to a government may have little to do with the risk of investing in a business. It is possible for a company to have a cost of equity that is lower than the interest rate on the government debt in the country. This seems counterintuitive, but compare the riskiness of a consumer packaged-goods (CPG) producer in an emerging market versus the government debt of that country. The CPG producer may experience a large drop in earnings during an economic crisis, but it typically springs back relatively quickly. And in contrast to the political environment facing banks and mining or energy companies, CPG businesses face little risk of appropriation by the government. With regard to government debt, however, it's not unusual for governments to default. Since 1990, Russia and Argentina have each defaulted, and oil-rich Nigeria has defaulted three times. Even Greece required bailout loans from the International Monetary Fund and European Central bank in 2010, 2012, and 2015. It's also possible that the cost of debt for some companies is lower than that of their government, as is the case in Brazil, where a number of companies' debt is rated investment grade while the government's is not.

Furthermore, it's illogical to apply the same risk premium across all industries. CPG producers generally survive economic disruptions, while banks may go bankrupt. For example, over the period 2013–2018, Brazilian ten-year government bonds were more volatile than beverage company Ambev and less volatile than the major Brazilian banks. Some companies (raw-materials exporters) might benefit from a currency devaluation, while others (raw-materials importers) will be damaged.

We've also found that the country risk premiums used in practice are too high and lead to overcompensation in the company's projected performance. Analysts using high premiums frequently compensate by making aggressive forecasts for growth and return on invested capital (ROIC). An example is the valuation we undertook of a large Brazilian chemicals company. Using a local weighted average cost of capital (WACC) of 10 percent, we reached an enterprise value of 4.0 to 4.5 times earnings before interest, taxes, depreciation, and amortization (EBITDA). A second adviser was asked to value the company

[2] T. Keck, E. Levengood, and A. Longfield, "Using Discounted Cash Flow Analysis in an International Setting: A Survey of Issues in Modeling the Cost of Capital," *Journal of Applied Corporate Finance* 11, no. 3 (1998): 82–99.

and came to a similar valuation—an EBITDA multiple of around 4.5—despite using a very high country risk premium of 11 percent on top of the WACC. The result was similar because the second adviser made performance assumptions that were far too aggressive: real sales growth of almost 10 percent per year and a ROIC increasing to 46 percent in the long term. Such long-term performance assumptions are unrealistic for a commodity-based, competitive industry such as chemicals. In another, broader set of analyst forecasts from 2015 to 2018, 30 percent of industries were expected to achieve growth rates more than 20 percent, while in the United States, only 5 percent were expected to achieve similar results. It's hard to imagine 30 percent of industries growing more than 20 percent per year.

These are among the reasons we favor a scenario DCF approach to valuing emerging-markets companies. It allows you to focus on company-specific risks, not generic risks.

Our empirical research also shows that there isn't much of a country risk premium built into the valuation of stocks in some emerging markets. If there were a substantial country risk premium, we'd expect price-to-earnings ratios (P/Es) to be much smaller than they are.

Consider Brazil. Over the past decade, many valuations we've seen have incorporated country risk premiums of 3 to 5 percent, plus an inflation differential versus U.S. companies of about 2 to 3 percent. That leads to a cost of equity of 15 to 18 percent. From 2015 to 2018, the P/E for the major Brazilian market index has been in the range of 10 to 17 times. Going back to the value driver formula derived in Chapter 3, we can solve for the expected growth in earnings, given estimates for the other values:

$$\frac{P}{E} = \left(1 - \frac{g}{\text{ROE}}\right) / (k_e - g)$$

where g is the growth rate of earnings, ROE is return on equity, and k_e is the cost of equity.

If we assume a P/E of 12 times, a cost of equity of 15 percent, and a marginal return on equity of 20 percent (above historical averages), the implied growth rate of earnings in perpetuity would have to be about 11.5 percent nominal, or about 7.5 percent in real terms (assuming 4 percent inflation, based on 2 percent in the United States and two percentage points higher inflation in Brazil). But 7.5 percent real growth in perpetuity is clearly unrealistic.

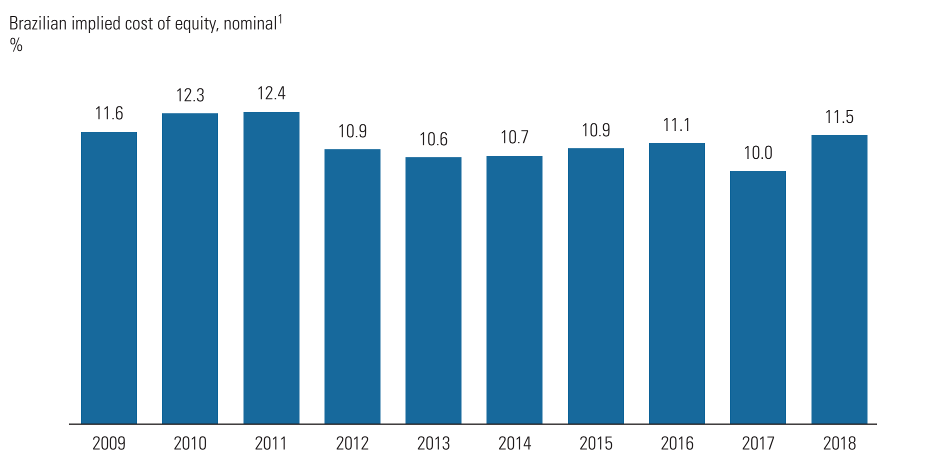
Looked at another way, if we assume 3.5 percent real growth in earnings in perpetuity (an optimistic view), the implied P/E at a 15 percent cost of equity is 8.3 times, which is about 30 percent lower than current P/Es. It's impossible to come up with a consistent set of assumptions that ties together a P/E of 12 and 15 percent cost of equity.

If we eliminate the country risk premium, our results work mathematically and economically. We'll use 2016 as an example and solve for the implied cost of equity. The P/E was about 13 times. Assuming 3.5 percent long-term real growth plus 4 percent inflation and a 14 percent ROE, we calculate a nominal cost of equity of about 11 percent. Subtracting inflation at 4 percent gives us a 7 percent real cost of equity, not very different from the real cost of equity of the United States (see Chapter 15).

Of course, these results are highly sensitive to small changes in some of the assumptions. The key point is that it is very difficult to reconcile current P/Es with a country risk premium of 3 percent or more. Exhibit 35.1 shows a time series of implied costs of equity for the Brazilian index over the ten years from 2009 to 2018. The nominal cost of equity stays within a range of 10 to 12 percent, or about 6 to 8 percent real. An implied country risk premium for Brazil is more likely to be closer to 1 percent than 3 to 5 percent.

One reason country risk premiums should be lower than levels used by many is that many country risks, including expropriation, devaluation, and war, are largely diversifiable. Consider the international consumer-goods player illustrated in Exhibit 35.2. Its returns on invested capital were highly volatile for individual emerging markets. Taken together, however, these markets were hardly more volatile than developed markets; the corporate portfolio diversified away most of the risks. Finance theory clearly indicates that the cost of capital should not reflect risk that can be diversified. This does not mean that diversifiable risk is irrelevant for a valuation: the possibility of adverse future events will affect the level of expected cash flows. But once this has been incorporated into the forecast for cash flows, there is no need for an additional markup of the cost of capital if the risk is diversifiable.

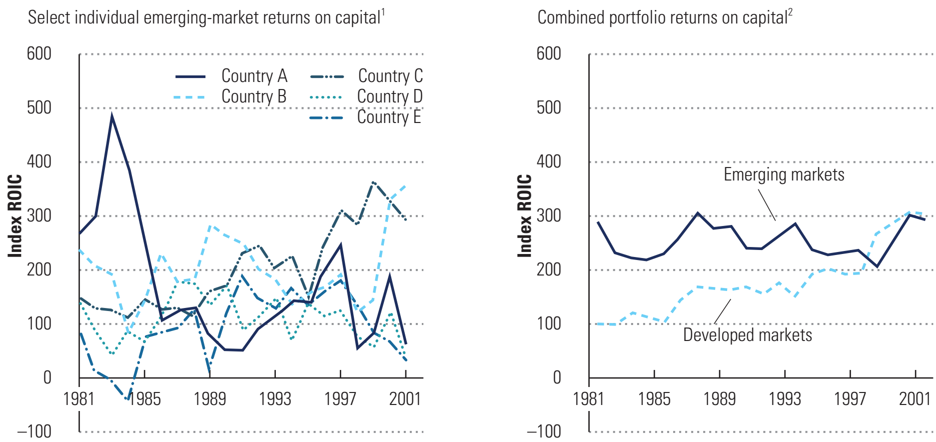
EXHIBIT 35.1 **Low Variability in Implied Cost of Equity**



[1] Assuming average prior 10-year ROE, 4.0% long-term inflation, 3.5% long-term real profit growth.

Source: Capital IQ.

EXHIBIT 35.2 **Returns on a Diverse Emerging-Market Portfolio**



[1] In stable currency and adjusted for local accounting differences.

[2] Combined portfolio included additional countries not reflected here.

Source: Company information.

Finally, most of us underestimate the impact that even a small country risk premium has on valuations, as we will show in the next section.

APPLYING THE SCENARIO DCF APPROACH

The preceding analysis of the Brazilian cost of equity masks a wide variation in P/Es across the economy. That's where the scenario DCF approach proves its advantages; it allows you to assess the risk of each company based on company-specific risk factors. At a minimum, model two scenarios. The first should assume that cash flow develops according to conditions reflecting business as usual (i.e., without major economic distress). The second should reflect cash flows assuming that one or more emerging-market risks materialize.

Exhibit 35.3 compares the valuation of a company with a European factory and an emerging-market factory with a similar outlook except for the emerging-market risk. In the example, the cash flows for the European factory grow steadily at 3 percent per year into perpetuity. For the factory in the emerging market, the cash flow growth is the same under a business-as-usual scenario, but there is a 25 percent probability of economic distress resulting in a cash flow that is 55 percent lower into perpetuity. The emerging-market risk is taken into account, not in the cost of capital but in the lower expected value of future cash flows from weighting both scenarios by the assumed probabilities. The resulting value of the emerging-market factory (€1,917) is clearly below the value of its European sister factory (€2,222), using a WACC of 7.5 percent.

EXHIBIT 35.3 **Scenario DCF vs. Country Risk Premium DCF**

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Net present value for identical facilities in . . .

Scenario approach

. . . a European market

Probability	Cash flows in perpetuity[1]	Year 1	2	3	4	. . .
100%	"As usual"	100	103	106	109	
0%	"Distressed"					
	Expected cash flows	100	103	106	109	
	Cost of capital			7.5%		
	Net present value		2,222			

. . . an emerging market

Probability	Cash flows in perpetuity[2]	Year 1	2	3	4	. . .
75%	"As usual"	100	103	106	109	
25%	"Distressed"	45	46	48	49	
	Expected cash flows	86	89	92	94	
	Cost of capital			7.5%		
	Net present value		1,917			86% of European NPV

Country risk premium approach

. . . a European market

Cash flows in perpetuity[1]	Year 1	2	3	4	. . .
"As usual"	100	103	106	109	
Cost of capital			7.5%		
Net present value		2,222			

. . . an emerging market

Cash flows in perpetuity[2]	Year 1	2	3	4	. . .
"As usual"	100	103	106	109	
Cost of capital			7.5%		
Country risk premium			0.7%		
Adjusted cost of capital			8.2%		
Net present value		1,917			86% of European NPV

[1] Assuming perpetuity cash flow growth of 3%.

[2] Assuming perpetuity cash flow growth of 3% and recovery under distress of 45% of cash flows "as usual."

We assumed for simplicity that if adverse economic conditions develop in the emerging market, they will do so in the first year of the plant's operation. In reality, of course, the investment will face a probability of domestic economic distress in each year of its lifetime. Modeling risk over time would require more complex calculations yet would not change the basic results. We also assumed that the emerging-market business would face significantly lower cash flows in a local crisis but not wind up entirely worthless.

We can also see from Exhibit 35.3 how easy it is to overestimate the country risk premium. As you can see, despite the 25 percent chance that the cash flows would be 55 percent lower than the base case, the equivalent country risk premium is only 0.7 percent (estimated by reverse engineering the valuation and solving for the discount rate based on the base-case cash flows). If we had used a country risk premium of 3 percent, the implied probability of economic distress would be 70 percent, versus 25 percent in the example.

Exhibit 35.4 gives an indication of the premium required for different combinations of the probability and size of an investment's permanent cash flow reduction. The premium is easily overestimated. For example, if there is a probability of 50 percent that future cash flows will be permanently lower by 40 percent, the risk premium should be just 1.5 percent. Actual premiums will also vary, depending on the underlying cash flow profile and cost of capital.[3] Nevertheless, the table allows for some calibration of premiums and risks.

While estimating probabilities of economic distress for the base case and downside scenarios is ultimately a matter of management judgment, there are indicators to suggest reasonable probabilities. Historical data on previous crises can give some indication of the frequency and severity of country

EXHIBIT 35.4 **Probability of Economic Distress Given Small Variations in Risk Premium**

Risk premium that reflects given conditions, %

Probability of lower cash flow, %	**Size of cash-flow reduction**, % 20	40	60	80	100
10	0.1	0.2	0.4	0.5	0.7
20	0.2	0.5	0.8	1.1	1.5
30	0.4	0.8	1.3	1.9	2.6
40	0.5	1.1	1.9	2.8	4.0
50	0.7	**1.5**	2.6	4.0	**6.0**

A 1.5% risk premium is assuming even odds that an investment will lose 40% of its value.

A 6% risk premium is assuming even odds it will lose all its value.

Note: Chart assumes a smooth cash-flow profile, 8% weighted average cost of capital, 2% terminal growth, binomial outcome.

Source: R. Davis, M. Goedhart, and T. Koller, "Avoiding a Risk Premium That Unnecessarily Kills Your Project," *McKinsey Quarterly* (August 2012).

[3] The higher the cash flow's growth rate, the stronger is the impact of a risk premium on the DCF value.

risk and the time required for recovery. We analyzed the changes in GDP of 20 emerging economies since 1985 and found that they had experienced economic distress, defined as a real-terms GDP decline of more than 5 percent, about once every five years. This would suggest a 20 percent probability for a downside scenario.

Another source of information for estimating probabilities is prospective data from current government bond prices.[4] Academic research suggests that government default probabilities in emerging markets such as Argentina five years into the future were around 30 percent in nondistress years.[5]

In our example, we could simply reverse engineer the country risk premium, because the true value of the plant was already known from the scenario approach. But for practical purposes, there is no agreed-upon approach to estimate the premium. Estimates from different analysts usually fall into a wide range because of the different methods used. The country risk premium is sometimes set at the so-called sovereign risk premium: the spread of the local government bond yield denominated in U.S. dollars and a U.S. government bond of similar maturity. However, that is reasonable only if the returns on local government debt are highly correlated with returns on corporate investments. In our experience, this is rarely the case.

From an operational viewpoint, using scenarios forces managers to discuss emerging-market risks and their effect on cash flows, thereby gaining more insights than they would secure from an arbitrary addition to the discount rate. By identifying specific factors with a large impact on value, managers can plan to mitigate these risks.

ESTIMATING COST OF CAPITAL IN EMERGING MARKETS

Calculating the cost of capital in any country can be challenging, but for emerging markets, the challenge is an order of magnitude greater. This section provides our fundamental assumptions, background on the important issues, and a practical way to estimate the components of the cost of capital.

General Guidelines

Our analysis adopts the perspective of a global investor—either a multinational company or a global investor with a diversified portfolio. Of course, some emerging markets are not yet well integrated with the global market. In China, for example, local investors may face barriers to investing outside

[4] See, for example, D. Duffie and K. Singleton, "Modeling Term Structures of Defaultable Bonds," *Review of Financial Studies* 12 (1999): 687–720; and R. Merton, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance* 29, no. 2 (1974): 449–470.

[5] See J. Merrick, "Crisis Dynamics of Implied Default Recovery Ratios: Evidence from Russia and Argentina," *Journal of Banking and Finance* 25, no. 10 (2001): 1921–1939.

their home market. As a result, local investors in such markets cannot always hold well-diversified portfolios, and their cost of capital may be considerably different from that of a global investor. Unfortunately, there is no established framework for estimating the capital cost for local investors. However, if the local stock market is fully integrated into the global markets (investors both in and out of the country can freely trade both locally and internationally), local prices will more likely be linked to an international cost of capital.

Another assumption is that, from the perspective of the global investor, most country risks are diversifiable. We therefore need no additional risk premiums in the cost of capital for the risks encountered in emerging markets when discounting expected cash flows in the scenario DCF approach. Of course, if you choose to discount the promised cash flow from the business-as-usual scenario only, you should add a separate country risk premium, as discussed earlier.

Given these assumptions, the cost of capital in emerging markets should generally be close to a global cost of capital adjusted for local inflation and capital structure. It is also useful to keep some general guidelines in mind:

- *Use the capital asset pricing model (CAPM) to estimate the cost of equity in emerging markets.* The CAPM may be a less robust model for the less integrated emerging markets, but there is no better alternative model today.
- *There is no one right answer, so be pragmatic.* In emerging markets, there are often significant gaps in information and data (for example, in estimating betas). Be flexible as you assemble the available information piece by piece to build the cost of capital.
- *Be sure monetary assumptions are consistent.* Ground your model in a common set of monetary assumptions to ensure that the cash flow forecasts and discount rate are consistent. If you are using local nominal cash flows, the cost of capital must reflect the local inflation rate embedded in the cash flow projections. For real-terms cash flows, subtract inflation from the nominal cost of capital.
- *Allow for changes in cost of capital.* The cost of capital in an emerging-market valuation may change, based on evolving inflation expectations, changes in a company's capital structure and cost of debt, or foreseeable reforms in the tax system. For example, in Argentina during the economic and monetary crisis of 2002, the short-term inflation rate was 30 percent. This could not have been a reasonable rate for a long-term cost of capital estimate, because such a crisis could not be expected to last forever.[6] In such cases, estimate the cost of capital on a year-by-year basis, following the underlying set of basic monetary assumptions.

[6] Annual consumer price inflation came down to around 5 percent in Argentina in 2004.

- *Don't mix approaches.* Use the cost of capital to discount the cash flows in a scenario DCF approach. Do not add any risk premium, because you would then be double-counting risk. If you are discounting only future cash flows in a business-as-usual scenario, add a risk premium to the discount rate.

Estimating the Cost of Equity

To estimate the components of the cost of equity, use the approach described in Chapter 15, with the following considerations for the risk-free rate, market risk premium, and beta.

In emerging markets, it is harder than in developed markets to estimate the risk-free rate from government bonds. Three main problems arise. First, most of the government debt in emerging markets is not, in fact, risk free: the ratings on much of this debt are often well below investment grade. Second, it is difficult to find long-term government bonds that are actively traded with sufficient liquidity. Finally, the long-term debt that is traded is often in U.S. dollars or the euro, so it is not appropriate for discounting local nominal cash flows.

We recommend a straightforward approach. Start with a risk-free rate based on the ten-year U.S. government bond yield, as in developed markets. Add to this the projected difference over time between U.S. and local inflation, to arrive at a nominal risk-free rate in local currency.[7] For emerging-market bonds with relatively low risk, you can derive this inflation differential from the spread between local bond yields denominated in local currency and those denominated in U.S. dollars.

Sometimes practitioners calculate beta relative to the local market index. This is not only inconsistent from the perspective of a global investor, but also potentially distorted by the fact that the index in an emerging market will rarely be representative of a diversified economy. Instead, estimate industry betas relative to a well-diversified or global market index, as recommended in Chapter 15.

Excess returns of local equity markets over local bond returns are not a good proxy for the market risk premium. This holds even more so for emerging markets, given the lack of diversification in the local equity market. Furthermore, the quality and the length of available data on equity and bond market returns usually make such data unsuitable for long-term estimates. To use a market risk premium that is consistent with the perspective of a global investor, use a global estimate (as discussed in Chapter 15) of 4.5 to 5.5 percent.

[7] Technically, we should also model the U.S. term structure of interest rates, but it will not make a large difference in the valuation.

Estimating the After-Tax Cost of Debt

In most emerging economies, there are no liquid markets for corporate bonds, so little or no market information is available to estimate the cost of debt. However, from a global investor's perspective, the cost of debt in local currency should simply equal the sum of the dollar (or euro) risk-free rate, the systematic part of the credit spread (which depends on the debt's beta; see the section titled "Estimating the After-Tax Cost of Debt in Chapter 15), and the inflation differential between local currency and dollars (or euros). Most of the country risk can be diversified away in a global bond portfolio. Therefore, the systematic part of the default risk is probably no larger than that of companies in international markets, and the cost of debt should not include a separate country risk premium.[8] Furthermore, companies in countries like Brazil often hold large amounts of cash to provide liquidity and minimize their net debt.

The marginal tax rate in emerging markets can be very different from the effective tax rate, which often includes investment tax credits, export tax credits, taxes, equity or dividend credits, and operating loss credits. Few of these arrangements provide a tax shield on interest expense, and only those few should be incorporated in the after-tax-cost-of-debt component of the WACC. Other taxes or credits should be modeled directly in the cash flows.

Estimating Capital Structure and WACC

Having estimated the cost of equity and after-tax cost of debt, we need debt and equity weights to derive an estimate of the weighted average cost of capital. In emerging markets, many companies have unusual capital structures compared with their international peers. One reason is, of course, the country risk: the possibility of macroeconomic distress makes companies more conservative in setting their leverage. Another reason could be anomalies in the local debt or equity markets. In the long run, when the anomalies are corrected, the companies should expect to develop a capital structure similar to that of their global competitors. You could forecast explicitly how the company evolves to a capital structure that is more like global standards. In that case, you should consider using the adjusted-present-value (APV) approach, discussed in Chapter 10.

OTHER COMPLICATIONS IN VALUING EMERGING-MARKETS COMPANIES

Other complications that should be considered in valuing emerging-markets companies include consistent macroeconomic parameters, accounting differences, nonoperating assets, and inefficient capital markets.

[8] This explains why multinationals with extensive emerging-market portfolios—companies such as Coca-Cola and Colgate-Palmolive—have a cost of debt that is no higher than that of their mainly U.S.-focused competitors.

Every forecast of a company's financial performance is based on assumptions about real GDP growth, inflation rates, interest and exchange rates, and whatever other parameters, such as energy prices, are relevant. In emerging markets, these parameters can fluctuate wildly from year to year. It becomes all the more important that forecasts be based on an integrated set of economic and monetary assumptions of future inflation, interest rates, exchange rates, and cost of capital (see Chapters 26 and 27 for more details). For instance, make sure that the same inflation rates underlie the financial projections and cost of capital estimates for the company.

One parameter deserves special attention: exchange rates. Although exchange rates converge to purchasing power parity (PPP) in the long run,[9] short-term deviations can be sizable and last for several years—especially in the case of emerging markets. In Chapter 27, Exhibit 27.3 shows how even on an inflation-adjusted basis, the exchange rate of Brazil's currency, the real (plural: reais), has fluctuated strongly over the past 50 years versus the U.S. dollar. If the long-term average real exchange rate is indicative of PPP,[10] the Brazilian currency could have been overvalued versus the U.S. dollar and other currencies by as much as 20 to 35 percent in 2008. Any exchange rate convergence to PPP would not be likely to affect the cash flows and value generated by a retailer, as its revenues and costs are mainly determined in Brazilian reais. But an exchange rate change would affect its cash flow and value measured in foreign currency. Because predicting exchange rates is virtually impossible,[11] a range estimate of the impact on a company's value measured in foreign currency is more meaningful. For primarily local companies, like retailers, it would therefore be best to perform the DCF valuation in Brazilian reais and—if needed—translate the result at both the actual and the PPP exchange rates to obtain a value range in foreign currency.

Fortunately, many of the complications arising from different accounting standards have been resolved over the past decades. Almost all countries outside the United States have adopted IFRS accounting standards, with the notable exceptions of China and India. This has reduced the complexity of adjusting their financial statements for valuation purposes. Even in China and India, the vast majority of accounting standards have been converging with IFRS and are now substantially the same.

Nonoperating assets remain a challenge, however. Companies in emerging markets—which are often conglomerates with a wide range of businesses—frequently have a large amount of nonoperating assets, including unconsolidated equity investments and real estate. For example, Reliance Industries,

[9] For an overview, see A. M. Taylor and M. P. Taylor, "The Purchasing Power Parity Debate," *Journal of Economic Perspectives* 18, no. 4 (Fall 2004): 135–158.

[10] See Chapter 27 for more details on PPP and exchange rates.

[11] As Exhibit 27.3 also shows, the Brazilian real further strengthened against the U.S. dollar and other currencies in real terms after 2008, before showing some correction in 2013.

one of India's largest companies, with 2018 revenues of $60 billion, has operations in oil refining and marketing, petrochemicals, oil and gas exploration and production, retail, digital services, and media and entertainment. Its balance sheet also includes $11 billion book value of investments that need to be valued separately (relative to a market capitalization of about $105 billion).

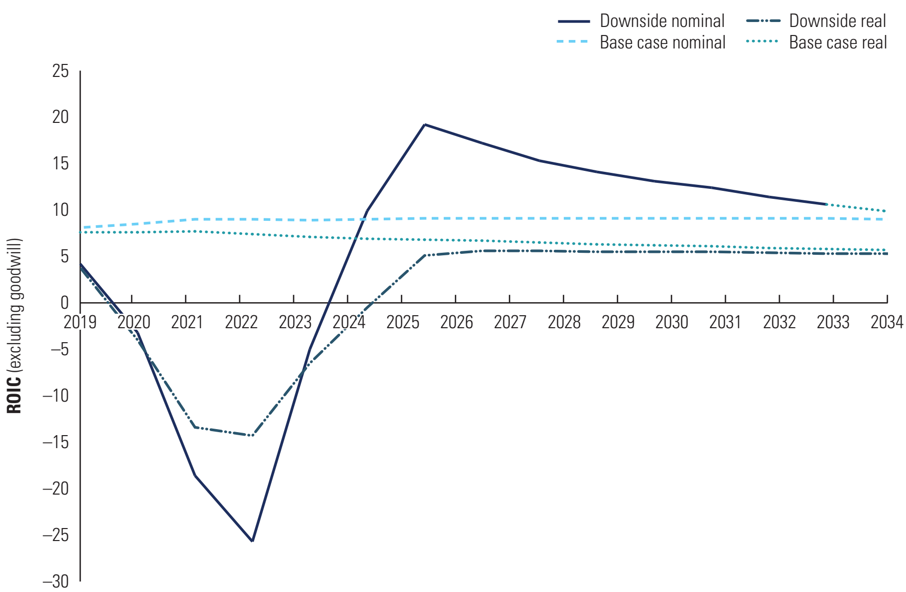
The capital markets in which emerging-markets companies trade may have inefficiencies. In many cases, these companies may have limited float because controlling shareholders may hold large stakes. The presence of controlling shareholders (often founding families) may also raise concerns about governance and whether there are potential conflicts between the interests of public shareholders and the controlling shareholders. This could lead to a lower share price than otherwise warranted. Some countries (particularly China and India) also have restrictions on investors, or the governments actively intervene in the markets, causing deviations in share prices from intrinsic values. For example, in China, Chinese citizens are not allowed to invest in shares outside the country, so the share prices of mainland Chinese companies can be disconnected from intrinsic value and the value of similar companies outside China. This could be caused by an imbalance of supply and demand for shares that cannot be corrected by arbitrage with other equity markets. Unlike most markets, the Chinese traded market is also dominated by retail investors (75 percent of holdings), roughly the reverse of the U.S. market, where institutional investors own most shares. Retail investors aren't as sophisticated and don't do as much research as institutional investors. They also tend to move in the same direction, leading to large swings in prices. Such market inefficiencies can make it difficult to reconcile DCF values with market values. Finally, companies in emerging markets often have complex corporate structures with voting and nonvoting shares. This often leads to a small group of investors controlling the company even though they own less than 50 percent. In some countries with weak governance, public market investors will discount the value of these companies if they don't believe the controlling shareholders make decisions in the interests of all shareholders.

TRIANGULATING VALUATION

We recommend triangulating the results of the scenario DCF approach with a comparable multiples approach and DCF using a country risk premium. We'll illustrate with the example of a Brazilian retail company we'll call ConsuCo.

We constructed two scenarios, a business-as-usual case (the base case) and a downside case reflecting performance under adverse economic conditions. Exhibit 35.5 shows the ROIC projections. Brazil has experienced several severe economic and monetary downturns, including an inflation rate that topped 2,000 percent in 1993. Judging by its key financial indicators, such as EBITDA to sales and real-terms sales growth, the impact on ConsuCo's business performance had been significant. ConsuCo's cash operating margin had been negative for four years, at around –10 to –5 percent, before recovering to its normal levels. In the same period, sales in real terms declined by 10 to 15 percent per year but grew sharply after the crisis. For the downside scenario projections, we assumed similar negative cash margins and a decline in sales, in real terms, for up to five years, followed by a gradual return to the long-term margins and growth assumed under the business-as-usual scenario.

EXHIBIT 35.5 **ConsuCo: ROIC and Financials, Base Case vs. Downside Scenario**



Financials, %	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Nominal indicators: Base case						
Sales growth	15.3	14.5	13.6	12.6	11.7	10.8
Adjusted EBITA/sales	6.1	6.2	6.4	6.4	6.4	6.4
NOPAT/sales	4.4	4.5	4.6	4.5	4.4	4.4
Invested capital (excluding goodwill)/sales	54	53	51	51	50	49
Invested capital (including goodwill)/sales	62	59	57	56	55	54
ROIC (excluding goodwill)	8.1	8.5	9.0	9.0	8.9	9.0
Free cash flow, reais million	(63)	(136)	(94)	(91)	(85)	113
Nominal indicators: Downside scenario						
Sales growth	10.0	25.0	66.3	66.3	25.0	11.3
Adjusted EBITA/sales	3.1	(2.2)	(8.0)	(7.6)	(1.1)	3.3
NOPAT/sales	2.3	(1.5)	(5.8)	(5.8)	(1.1)	2.2
Invested capital (excluding goodwill)/sales	55	47	31	22	21	22
Invested capital (including goodwill)/sales	63	54	35	25	23	24
ROIC (excluding goodwill)	4.2	(3.2)	(18.6)	(25.7)	(5.0)	9.9
Free cash flow, reais million	(149)	(777)	(2,533)	(4,504)	(2,677)	(558)

We discounted the free cash flows for ConsuCo under the base case and the downside scenario. The resulting present values of operations are shown in Exhibit 35.6. Note that we conducted the analysis in both nominal and real cash flows to show that the results were identical. We then weighted the valuation results by the scenario probabilities to derive the present value of operations. Finally, we added the market value of the nonoperating assets and subtracted the financial claims to arrive at the estimated equity value. The estimated equity value obtained for ConsuCo was about 32 reais per share, given a 30 percent probability of economic distress. This was somewhat lower than ConsuCo's share price in the stock market of around 37 reais at the time of valuation.

To triangulate with multiples, we apply Chapter 18's guidance on how to perform a best-practice multiples analysis to check valuation results. For the ConsuCo example, we compared the implied multiple of enterprise value over EBITDA with those of peer companies. All multiples were forward-looking

EXHIBIT 35.6 **ConsuCo: Scenario DCF Valuation**

reais, million

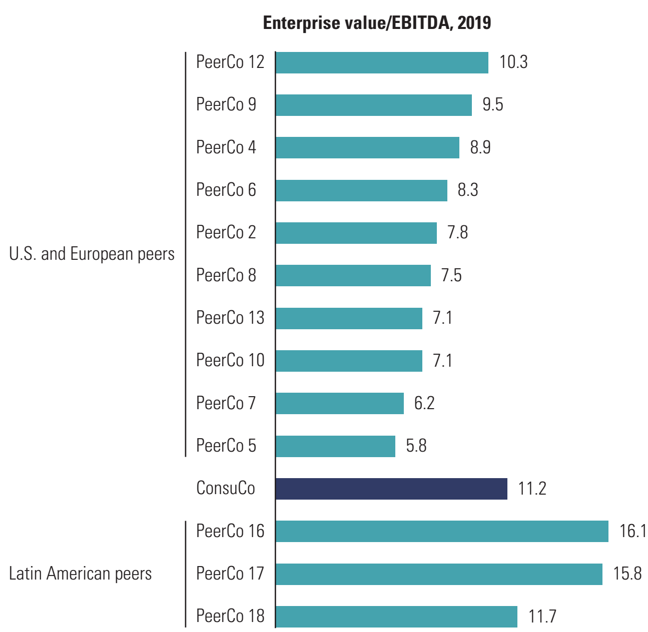
		2019	2020	2021	2022	2023	2024	. . .	2029	. . .	2034
	Base case										
	Nominal projections										
	Free cash flow	(63)	(136)	(94)	(91)	(85)	113	. . .	301	. . .	516
	WACC, %	11.1	9.5	9.3	9.2	9.1	9.0	. . .	9.0	. . .	9.0
	Real projections										
	Free cash flow	(60)	(125)	(83)	(77)	(68)	87	. . .	187	. . .	257
	WACC, %	6.0	5.1	4.9	4.7	4.5	4.4	. . .	4.4	. . .	4.4
	DCF value	14,451									
	Nonoperating assets	1,139									
Probability 70%	Debt and debt equivalents	(5,605)									
	Equity value	9,985									
	Value per share	42.4									
	Downside scenario										
	Nominal projections										
Value per share 32	Free cash flow	(149)	(777)	(2,533)	(4,504)	(2,677)	(558)	. . .	250	. . .	834
	WACC, %	11.1	29.4	76.7	76.4	28.7	9.5	. . .	9.0	. . .	9.0
	Real projections										
	Free cash flow	(142)	(593)	(1,105)	(1,123)	(534)	(106)	. . .	38	. . .	102
	WACC, %	6.0	3.5	1.0	0.8	2.9	4.3	. . .	4.4	. . .	4.4
	DCF value	6,313									
	Nonoperating assets	1,139									
Probability 30%	Debt and debt equivalents	(5,605)									
	Equity value	1,847									
	Value per share	7.9									

multiples over EBITDA. As Exhibit 35.7 illustrates, the implied multiple from our ConsuCo valuation was significantly higher than for U.S. and European peers, which was not surprising, given its higher growth outlook in the Brazilian market compared with that of large established chains in the U.S. and European markets. ConsuCo's valuation was at the low end of the range for Latin American peers, which also was not unreasonable. Relative to regional peers, ConsuCo could have been expected to have fewer growth opportunities, as it was already very well established and geographically widespread. It also had somewhat more exposure than listed peers had to the lower-growth food segment.

The last part of the triangulation consisted of valuing ConsuCo using a country risk premium approach. Using Exhibit 35.4, we estimated a country risk premium for ConsuCo. We observed from history that the probability of country crises appears to be around 20 to 30 percent and that for consumer goods businesses, it rarely leads to a loss of all cash flows. Taking that into account, a country risk premium for a Brazilian retailer like ConsuCo was likely in the range of 1 to 2 percent, rather than 3 to 5 percent or higher, as analysts often estimate.

Discounting the business-as-usual scenario at the cost of capital plus a country risk premium in this range led to a value per share below 20 reais, far lower than the 32-reais result obtained in the scenario DCF approach. The

EXHIBIT 35.7 **ConsuCo: Multiples Analysis vs. Peers**



reason for this gap lies in ConsuCo's cash flow profile, and it highlights why a scenario approach is preferable to using a discount rate reflecting a country risk premium. Due to ConsuCo's high anticipated growth and corresponding investments, its free cash flows were forecast to be negative for the first five years, pushing value creation forward in time. But the further ahead a company's positive cash flows lie, the more those cash flows are penalized by the country risk premium approach, because a markup in WACC accumulates over time. This does not happen in a scenario approach, because the scenario probabilities affect all future cash flows equally.

If ConsuCo had had a lower-growth outlook, the country risk premium approach would have produced a valuation much closer to the valuation from the scenario approach. Note that irrespective of ConsuCo's cash flow profile, a risk premium of 3 to 5 percent (as is typically used in emerging markets) would have either resulted in unrealistically low valuations relative to current share price and peer group multiples or else required an unrealistically bullish forecast of future performance.

SUMMARY

To value companies in emerging markets, we use concepts similar to the ones applied to developed markets. However, it's necessary to incorporate into valuations the unique risks of emerging markets, such as macroeconomic or political crises, by following the scenario DCF approach. This approach develops alternative scenarios for future cash flows, discounts the cash flows at the cost of capital without a country risk premium, and then weights the DCF values by the scenario probabilities. The cost of capital estimates for emerging markets build on the assumption of a global risk-free rate, market risk premium, and beta, following guidelines similar to those used for developed markets. Since company values in emerging markets are often more volatile than values in developed markets, we recommend triangulating the scenario DCF results with two other valuations: one that is based on discounting cash flows developed in a business-as-usual projection but using a cost of capital that includes a country risk premium, and another that is based on multiples.

REVIEW QUESTIONS

1. Define *purchasing power parity*. What is the importance of purchasing power parity when you are trying to establish value for a company located in an emerging market?
2. Identify four risks associated with emerging markets that affect enterprise discounted-cash-flow (DCF) valuation. How should these risks be treated within the enterprise DCF model?
3. Describe the benefits of a scenario DCF valuation model. What factors should be considered when constructing scenario parameters?
4. You are computing the value of a firm headquartered in an emerging market. Identify the factors unique to an emerging market that need to be evaluated when estimating the cost of equity using the capital asset pricing model (CAPM).
5. Volatilities for individual stock and market indexes in emerging economies are typically higher than those for U.S. stocks and indexes. Should that mean that the cost of capital for investments in emerging economies is higher, too? Explain your answer.
6. Discuss the relative merits of including risk adjustments in cash flow or in discount rates—especially for high-growth companies in emerging markets.
7. To estimate the beta for a Brazilian telecommunications company, you have collected a sample of telecom peers in Latin America and Asia and peers in the United States and Europe. The median beta versus a world index is around 1.5 for the Latin American and Asian peers and around 0.9 for the U.S. and European peers. What would you have to believe to choose the Latin American/Asian peer beta? What additional analyses would you undertake to test either choice?
8. Many emerging economies have restrictions on capital outflows to protect their growth and stability; for example, they may impose high taxes on repatriated profits by foreign companies. Where and how would you include such taxes in the DCF valuation of your company's subsidiary in a high-growth emerging economy, if the taxes are (a) levied in perpetuity or (b) gradually decreased to zero over the next ten years as the economy starts to mature?